Lucas Domingues, MSc | CMIOSH

Become a SAFETY 4.0 Leader



"A must read for anyone seeking to future-proof their career and become a safety leader in the digital age"

2026 Edition

BECOME A

Safety 4.0 Leader

Don't just keep up with change. Lead it.

Lucas Domingues
MSc, CMIOSH

Become the Safety 4.0 Leader

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Safety 4.0 Academy

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Dedicated to my lovely wife Mariana and our two beautiful boys —
Thomas & Martin. Dad loves you.

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About the Author



Lucas Domingues, MSc, CMIOSH, is the founder and CEO of Shield360 and the Safety 4.0 Academy. He is also a podcaster and creator of Safety 4.0 Live — a meeting point for safety leadership, innovation and impact.

With 15+ years of experience spanning energy, construction, and technology, Lucas has led safety innovation at both startup and enterprise levels. He created the Safety 4.0 Academy to help safety professionals worldwide adapt, grow, and lead in the digital age.

Let's connect on LinkedIn.

Foreword

Prakash Senghani | *CEO, Navatech*

I first came across Lucas when he reached out to me on LinkedIn to introduce himself and talk about safety innovation. He later invited me to join him as the first guest on his podcast, which I gladly accepted. Since then, we've kept in touch, and over time I've come to see him not just as a colleague but as someone I've been able to share advice and perspectives with, almost in the role of a mentor.

It's not about replacing what you know; it's about building on it with curiosity, innovation, and practical action.

When I was working on site, I often noticed how safety observations were captured in the most informal ways — scribbled into notepads, shared quickly in group chats, or mentioned in passing on the job. What struck me wasn't that people weren't paying attention — they were. They cared enough to notice and to speak up. But the way those observations made their way into formal systems felt clumsy. Safety leaders would spend hours copying notes into databases, trying to track them and spot patterns before something went wrong.

The fact that you've picked up this eBook tells me you're already on that journey. You're curious, open to new ideas, and willing to think differently about safety. That's an excellent start. But curiosity must turn into action. My encouragement is simple: as you read, don't just absorb the principles — apply them. Try out one idea in your next team meeting, test a new approach to incident reporting, or encourage your people to share observations in ways that feel natural to them. Small steps can unlock big changes.

And remember, this isn't a solo effort. The real transformation begins when leaders create space for their teams to be inquisitive too — when asking questions, challenging assumptions, and spotting patterns becomes part of everyday work. That's when safety stops being a box to tick and starts becoming a shared way of thinking.

This book is an invitation — to lead with curiosity, to take practical steps, and to nurture a culture where every observation, no matter how small, can spark learning and change. Because the leaders who ask better questions will always discover better answers.



Preface

This eBook is written for two kinds of safety people. First, for safety leaders who are already carrying the responsibility of protecting people and organisations, but who now face the challenge of leading in the digital age. The truth is, none of us were taught this at the safety school or on the job. No one prepared us for a world where safety leadership would mean understanding data, using AI responsibly, or adapting to new technologies as fast as they appear. Yet this is the reality we must lead in today.

Second, it's for safety professionals who want to become the leaders of tomorrow. Those who see that the role is no longer about compliance and checklists, but about building trust, using technology wisely, and shaping culture in real time. If that's you, this eBook will give you the perspective and the tools to accelerate your journey.

What you'll find here is not theory or jargon. It's a personal guide, drawn from my own experiences — from shop floors to boardrooms — and from seeing firsthand how safety leadership has changed in the last twenty years and recent months. Inside, we'll look at why the old approaches no longer work, how to escape the reactive safety trap, what new skills and tools matter most, and how to build a roadmap that makes safety more proactive, human, and connected.

Above all, this book is here to give you confidence. Confidence to step into Safety 4.0, to use digital tools without fear, and to lead with clarity in times of change. Because the future of safety will not be written by those who wait for permission — it will be shaped by those who are ready to act.

— Lucas Domingues

Chapter 1

Safety Leadership Reimagined



"Safety leadership changed more in the last few months than in the last two decades"

When I look back at my own career, the role of a safety leader has changed more in the last year than in the twenty before. I started as you probably did — surrounded by checklists, clipboards, and rules that often felt like they were written in stone. Leadership then was about enforcement and expertise. You knew the regulations, you kept people compliant, and you were respected for being the person who could quote the law and spot the hazard.

But as I moved from the shop floor to the boardroom, I realised that leadership in safety is no longer about being the rule-keeper. It's about being the sense-maker and change-maker. In today's world, change happens in real time. Technology advances faster than regulation. Information floods in from sensors, systems, and workers in ways we never imagined before.

That puts a big question mark over what safety leadership really means. The old style of authority — top-down control, technical dominance, and slow decision cycles — simply doesn't fit in a world where risks evolve daily, and technology can outpace human adaptation. So, what does modern safety leadership look like?

Power Skills Over Hard Skills

Technical expertise will always matter, but it's no longer enough to carry influence. The leaders who thrive now are those with power skills — communication, empathy, storytelling, influence, and the ability to create alignment across disciplines. These aren't "soft skills." They're the hardest to master and the most impactful in building a culture where safety is seen as everyone's responsibility.

Adaptive Intelligence at the Core

In the past, leaders could rely on experience to guide them. Today, experience helps, but adaptability matters more. Adaptive intelligence means you can make sense of fast-changing situations, adjust course without hesitation, and learn faster than the pace of change. It's about having a growth mindset in a world where yesterday's solution might be obsolete tomorrow.

Critical Thinking in the Age of AI

AI brings speed, but speed without judgment can be dangerous. Leadership now requires critical thinking more than ever — asking why an algorithm is making a certain recommendation, how a model is trained, and what blind spots might remain. Workers will trust AI only if they see their leaders applying wisdom, not blind faith, to its outputs.

The Safety 4.0 Leader

The Safety 4.0 leader won't just manage compliance. They'll shape culture. They'll use technology not as a crutch, but as a lever. They'll combine power skills with adaptive intelligence, balancing human connection with digital insight. And most importantly, they'll move fluidly between the factory floor, worksite and the boardroom, because real leadership lives in both places.

> **Key Takeaway:** *Modern safety leadership is no longer about being the rule-keeper — it's about becoming a sense-maker and change-maker, blending power skills, adaptive intelligence, and critical thinking to lead in real time.*

Chapter 2

Why I Wrote This eBook?



"We weren't trained at the safety school or on the job for the future ahead of us"

I didn't set out to become "the Safety 4.0 guy." I started where most of us start: on the ground, steel-toed boots, hi-vis vest and an A4 clipboard, chasing signatures, delivering briefings and investigating what went wrong. I remember sweating through inductions in the sugarcane fields under the hot sun in Brazil, then working around the clock to prepare the paperwork for the next audit and occasionally making PowerPoint slides showing the lagging indicators triangle (yes, Bird's pyramid) as we literally chased KPIs set by someone in the leadership team. It felt important as I was always busy, but it also felt like we were forever tidying yesterday and never winning.

I cannot complain about my early days at all as it was a period of deep learning, stretching me out of the comfort zone and establishing the foundation for becoming the Lucas of today. I am grateful for each opportunity, every struggle as well as small and big wins along the way.



Presenting a toolbox talk to rural workers in Brazil (2007)

From my humble beginning in the sugarcane fields to having a 360° career transformation in about 2 years' time when I joined an international offshore drilling operator in Rio de Janeiro called Sevan Drilling.



Standing in front of a BOP — blowout preventer (2009)

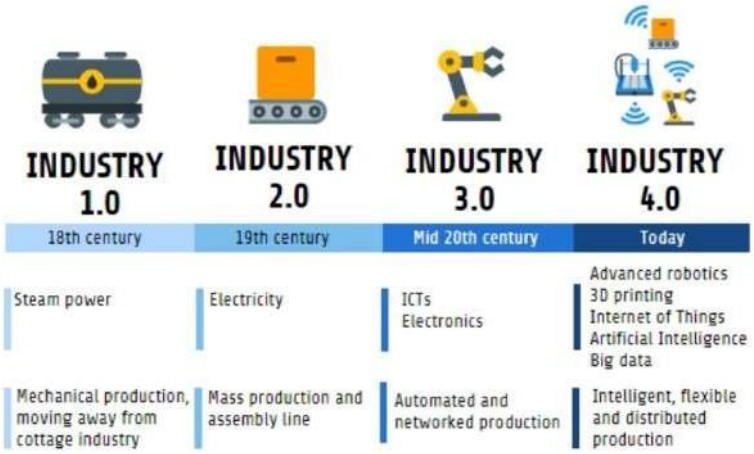
But I felt something was missing and safety felt boring for the first time, even in such a dynamic work environment. I hit the wall of bureaucracy, excessive admin work and over-reliance on paperwork. For a long time, I thought maybe I was thinking too far ahead. Then, slowly things started to change and some innovations were introduced — something like the "state of the art EHS software" or a fancy document management system. I am pretty sure that if I hadn't worked for over 10 years in the energy industry that provided me so many opportunities, I would probably have quit safety as a profession.

Then came the day I wished would never come and that made me stay. Multiple contractors, tight drilling program, weather turning, radios hissing with the kind of static that makes everyone talk louder. We had checklists, permits, briefings. We also had near misses stacking up like dominoes. I kept thinking: we don't have a culture problem; we have a visibility problem. And then — a fatality.

I will always remember that day. A nice young roughneck, safety-conscious and fully trained, fell from height — a tragic death. A few years later, another fatality onboard. We know how devastating it is.

I've spent years straddling two worlds: the practical one of boots, checklists and barriers, and the strategic one of boardrooms and budgets. I've worked in startups where everything moves before the ink dries, and in established organisations where inertia has its own parking space. In both, I saw the same patterns: brilliant people trapped in reactive systems; good intent diluted by bad information; safety measured by paperwork velocity rather than risk control. We were busy. We weren't always effective.

Meanwhile, the rest of the business was changing. Operations plugged into real-time data. Maintenance became predictive. Supply chains learned to learn. Industry 4.0 arrived with connected machines and smarter decisions, yet safety often stayed analogue in a digital world. We digitised forms but not thinking. We bought tools but didn't change habits. We tried to fix tomorrow with yesterday's tools.



Industry 4.0 timeline

Safety 4.0, for me, isn't a slogan or a shopping list. It's a shift in how we lead. It says: let's make safety as connected as the work it protects. Let's move from finding out to finding out early. Let's use the tools we already have — data, simple automation, better communication — to free people to do the work only people can do: notice, decide, influence, care.

Culture is not a poster; it's a thousand daily choices made easier or harder by the system around them. Processes either create flow or create friction. Technology either disappears into the work or becomes another thing to manage. And underneath it all is data — not in the abstract, but as the quiet thread that ties conversations to actions, actions to outcomes, and outcomes back to learning.

This eBook is personal. It's built from projects that went right and a few that taught me more than I asked for. It's about the moments when a supervisor texts: "We fixed it before it broke," or when a new starter speaks up because the reporting path is obvious and safe. It's about replacing a weekly chase with a quiet automation; swapping a four-page memo for a two-minute video that people actually watch; turning "we should have known" into "we saw it coming."

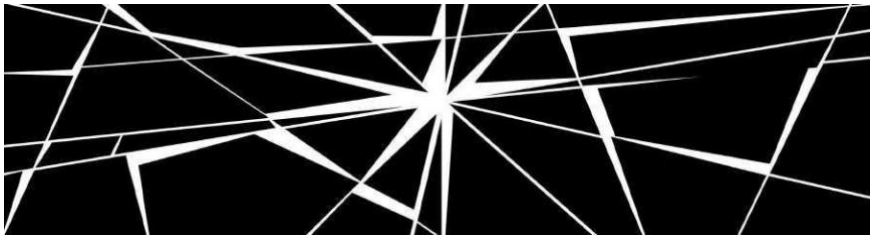
So, who is this for? Here's what you can expect from the chapters ahead. First, we'll tell the truth about the status quo — the firefighting that looks like progress until it doesn't. Then we'll define what Safety 4.0 is — not as an abstract framework but as a way of working that joins people, process, and technology with data as the quiet foundation. We'll look at the skills modern safety leaders actually need, and why you don't have to become a technologist to be effective. We'll get practical with low-effort, low-cost changes that create momentum.

If you're expecting a manifesto about replacing people with AI, this isn't it. If you're ready to elevate your career profile and become the Safety 4.0 leader — one smart step at a time — you're in the right place. Let's begin.

Key Takeaway *This book was written because I noticed a gap that keeps widening. The skills we used to rely upon are changing. Modern safety leaders know they need to learn, adapt and thrive. Keeping up is no longer enough. We must lead this change.*

Chapter 3

The Safety Status Quo is Broken



"And it's costing more than you think."

For all the progress we've made in safety over the last few decades, I still see the same story play out in too many organisations. The posters get updated. The reporting templates change colour. A new software tool appears. But the underlying reality is the same game, with the same rules, and the same blind spots. We're still measuring success by the absence of bad news. We're still treating safety as a compliance exercise, where the main objective is passing the next audit. And we're still spending far too much time reacting to yesterday's problems instead of preventing tomorrow's.

I call this the reactive safety trap — and it's the default setting in many workplaces. When you're in it, you may not feel it. Your calendar is too busy and you're always working towards fixing something broken. One phone call can wipe out your week. You spend more energy on closing investigation reports than actually fixing the underlying issues. The trap has many moving parts. It starts with data silos and fragmentation — inspection records in one system, incident logs in another, and near-miss reports buried in someone's inbox or Excel spreadsheets.



The reactive safety trap

The signals are there, but they're scattered, so no one sees the full picture in time. Add to that an excessive administrative workload, where safety professionals spend hours in front of

computers re-entering the same information into different forms. Layer in the firefighting cycle, where urgent issues always push planned improvements to "next week." Then there's low digital literacy, which means even when tools exist, people don't use them to their full potential.

And above all, senior leadership remains obsessed with lagging indicators like TRIR, LTI rates, or the neat ratios in Bird's Triangle.



Bird's Triangle model

Yes — the model created in 1932 that still seems to influence board directors today. These numbers look nice on a slide deck, but they only tell you how you performed in the past, not what's about to happen. My personal resistance isn't about the age of the model. My proposal isn't about scrapping Bird's Triangle — it's about being open-minded about other ways of measuring impact and performance using proactive data.

When you're stuck in this loop, it can feel strangely safe. At least you know what to expect: something will happen, you'll respond, and the cycle will repeat. But it's an illusion. Being good at reacting doesn't mean you're in control — it just means you've learned to clean up well. And clean-up is always more expensive, more disruptive, and more damaging than prevention.

I've seen organisations break free from this trap. The turning point comes when they stop seeing safety as a box-ticking exercise and start treating it as a core part of performance and business strategy. They track leading indicators. They share information in real time. They encourage people to report the almosts and the nearlys without fear. They measure whether fixes actually work, not just whether they're "closed."

This isn't about flooding teams with fancy tech or chasing the latest buzzwords. It's about designing a system where the signals come early, the data flows freely, and people have both the skills and the confidence to act. That's what Safety 4.0 promises.

Key Takeaway: *As long as we stay trapped in data silos, paperwork, and lagging indicators, safety will remain reactive instead of transformative.*

Chapter 4

The Dawn of Safety 4.0



"It can be hard to see the full shape of the day ahead."

The first time I came across the term Industry 4.0 was over a decade ago. Back then, it was being used to describe what many called the "Fourth Industrial Revolution" — a powerful convergence of advanced manufacturing, automation, robotics, data analytics, and machine learning that was redefining how products were designed, built, and delivered.

In simple terms, the industrial journey began with Industry 1.0, when steam power and mechanisation replaced manual labour. Industry 2.0 brought electricity, assembly lines, and the birth of modern manufacturing. Industry 3.0 arrived with electronics, computing, and early automation. Now, Industry 4.0 represents the fusion of the physical and digital worlds — smart factories, connected machines, predictive analytics, and artificial intelligence working together.

As industries evolved through these revolutions, so too did the nature of workplace risks and the methods we use to manage them. Safety has always followed industry, adapting to the risks of the era. But as technology was being adopted to power industry, safety seemed to be left behind — machines becoming intelligent while the systems protecting the people running them were stuck in the past with paper forms, siloed reports, and reactive thinking. That gap is where the idea of Safety 4.0 was born.

The earliest conversations about Safety 4.0 were happening in pockets of forward-thinking safety teams in manufacturing, energy, and construction — experimenting with IoT sensors, AI-driven risk analysis, and digital workflows. But there was no shared language for what they were doing. The term Safety 4.0 gave it that language. It positioned safety not as a lagging part of transformation, but as an integral driver of it.

I remember visiting a 6th generation drill ship in the Gulf of Mexico that had embraced the concept of "Rig of the Future." At the operational heart of the ship, a revolutionary red zone management system powered by computer vision had been implemented.



Smart red zone management powered by computer vision

The Red Zone monitoring solution constantly monitors people and equipment in the Red Zone. Robust and easy to implement, the system has cameras constantly overseeing operations and an intuitive interface to alert the driller when people enter the Red Zone. It wasn't flashy tech for tech's sake; it was smart, invisible, and deeply human-centred.

Another example comes from an EV manufacturing startup in London I worked with between 2020 and 2022. The microfactory robotic cells had autonomous mobile robots called WeMos capable of carrying payloads of up to 1 ton. These robots utilised a combination of IoT sensors to perceive their environment and navigate safely, even in dynamic settings with workers around.



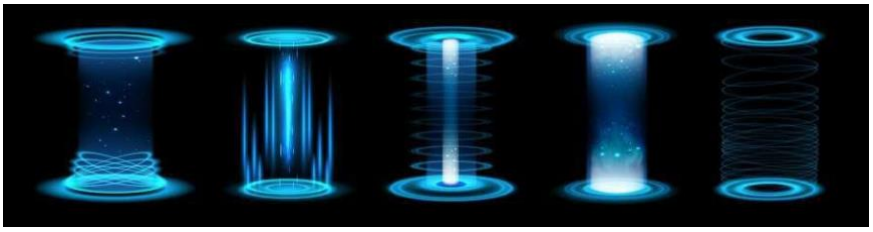
Arrival's WeMo autonomous robots

We're at the dawn of this shift. And like any dawn, it can be hard to see the full shape of the day ahead. Many organisations are still in the Safety 2.0 or 3.0 era — heavily manual, reactive, and compliance-driven. Others are experimenting with technology, but without the cultural or strategic alignment to make it stick. Safety 4.0 isn't a magic button. It's a journey. And it starts with understanding that technology is not the end goal — safer, healthier, more engaged people are. Tech is simply the bridge.

Key Takeaway *Safety 4.0 is not about tech for tech's sake — it's about using data as a bridge to connect people, culture, and process for smarter decisions and safer outcomes.*

Chapter 5

Pillars of Safety 4.0



"A strong foundation doesn't need to be complicated."

When people talk about "digital transformation" in safety, the first thing that comes to mind is often technology — the new tools, the smart devices, the AI dashboards. But I've learned, sometimes the hard way, that technology on its own changes very little. You can buy the most advanced system in the market, but if your people don't trust it, your culture doesn't support it, your processes can't absorb it, and your data is a mess, you've just added another shiny layer to an outdated structure.

That's why I see Safety 4.0 as being built on three interconnected pillars: people and culture, processes, and technology — supported by data as a foundational requirement. If one of them is weak or missing, the whole thing wobbles.



Safety 4.0 pillars

People and Culture

I combine these two because you can't separate them in practice. People bring skills, knowledge, and attitudes. Culture is the collective behaviour and unwritten rules that guide how those people act. In a Safety 4.0 environment, the goal isn't to replace people with machines — it's to empower them. That means building trust in the tools, giving people the skills to use them, and fostering a culture where digital solutions are seen as allies, not threats. When people feel ownership of safety, they're more likely to embrace change.

Process

Processes are the operating system of any safety function. They define how information flows, how decisions are made, and how actions are tracked. Too often, processes in safety are designed for compliance — to meet a regulation, satisfy an auditor, or pass a certification. In Safety 4.0, processes are designed for agility. They're still compliant, but they're built to integrate new tools, handle data in real time, and support proactive decision-making.

Technology

This is the visible part of the structure — IoT sensors, wearables, mobile apps, drones, computer vision, AI-driven analytics. These tools can radically change how safety is managed, but only if they're chosen for the right reasons and implemented with the other pillars in mind. I've also seen teams with little or no budget using the tools available in their organisations to build impressive digital systems using Microsoft or Google apps. In Safety 4.0, technology isn't about ticking an "innovation" box; it's about solving real problems.

Data — The Foundational Pillar

Data is what underpins everything else. Without good data, people are making decisions in the dark, processes are based on guesswork, and technology is just a collection of disconnected gadgets. In Safety 4.0, data flows freely across systems and teams, is collected in real time, cleaned, and organised so it can actually be used. It's not just numbers on a dashboard; it's context-rich information that helps predict and prevent harm.

And here's the important part: you don't need to wait for a big budget or executive sign-off to start. Many powerful tools including AI are now free or low-cost. Often, the solution you need is sitting with the team next door, in IT, operations, or even HR. The real driver is strategy: knowing where you are, where you want to be, and how to connect the dots using the resources already within reach.

> **Key Takeaway** *People, process, and technology matter — but it's data that underpins them all and turns safety into a living system.*

Chapter 6

The Greatest Skills Shift



"The marathon runner analogy."

When I first started in safety, the role was defined by expertise and experience. You earned your place by knowing the regulations inside out, by being able to quote clauses from memory, and by having the experience to back it up. If you could walk onto a site, spot a non-conformance from distance, and explain exactly which legal requirement it breached — you were valuable. That hasn't changed — deep knowledge of compliance still matters. But here's my point of view: it's no longer enough.

In this analogy, the robot and the racing car represent technology, AI, and modern tools. And this is where your choice becomes clear: you can complain about the new rules and keep running the old way, or you can adapt, step into the fast car, and take the lead in shaping the future.



The marathon runner analogy

Think about it like navigation. In the past, you might have been the person who could read the paper map better than anyone else. That skill was critical. But now, GPS exists. Your value isn't in fighting to keep the old map; it's in understanding how to program the GPS, interpret the data it gives you, and decide the best route when conditions change.

And conditions are changing. The World Economic Forum's Future of Jobs Report 2025 makes it clear: **nearly half of all workers will need reskilling between 2025–2030.**

For safety professionals, that's a wake-up call. It's not about turning you into a coder or an AI prompt engineer — it's about building the ability to work alongside technology, interpret its outputs, and lead people through change. This is where adaptive intelligence becomes essential: the ability to adjust quickly to new information, environments, and challenges.

In practical terms, this skills shift means: moving from purely compliance-driven thinking to strategic risk leadership; translating technical data into human stories that motivate action; knowing enough about AI, IoT, automation and data analytics to see opportunities and avoid pitfalls; being comfortable with ambiguity and change, even when the roadmap isn't clear; and building influence without relying on positional authority.

I've been working with safety leaders who embraced this shift, and the difference is striking. They don't see new tools as a threat; they see them as force multipliers. They use AI to draft reports in minutes instead of hours, freeing time for high-value conversations in the field. They bring together IT, operations, and HR to solve problems collaboratively.

The resources are more accessible than ever. You will find amazing free courses by Google, OpenAI, and Microsoft. Example: academy.openai.com

> **Key Takeaway** *You don't need to be a tech expert to thrive — you need digital literacy, adaptive intelligence, and the power to influence change.*

Chapter 7

Your Safety 4.0 Roadmap



"No roadmap is perfect. Just start."

By now, you've seen why the old ways aren't enough, what Safety 4.0 really means, and the skills and tools that can make it real. But the question most safety professionals ask is: Where do I actually begin? The truth is, transformation doesn't happen through grand strategies or big budgets. It happens in small, deliberate steps that build confidence and momentum. The key is simple: start small, act fast, and let progress compound.

Step One: Begin with Honesty

Every journey starts by taking a hard look at where you are today. Maybe your team is still buried in paper forms. Maybe your systems don't talk to each other. Maybe senior leaders are still obsessed with lagging indicators like TRIR or Bird's Triangle. That's okay. You can only start from where you are. My honest recommendation is to take the

Safety 4.0 Readiness Scorecard — it's a quick way to map your starting point and see which gaps matter most.

Step Two: Pick One High-Friction Process

Choose one high-friction process — maybe incident reporting, action tracking, or toolbox talk attendance — and make it easier with a tool you already have. It could be Microsoft Forms, Google Workspace, or even a simple QR code system. When you deliver a visible improvement, share it widely. Celebrate it. These early wins prove that Safety 4.0 isn't theory — it's a better way of working.

Step Three: Scale with Evidence

Once information flows digitally and you have proven results, suddenly you can spot trends, act early, and demonstrate the value of proactive and predictive safety. This is the turning point, when Safety 4.0 stops being a pilot project and becomes part of the way your organisation operates.

Step Four: Keep It Human, Keep It Moving

Your roadmap doesn't need to be perfect — it needs to be practical. It should grow with your people, your culture, and your business. Progress always beats perfection. The leaders who thrive in Safety 4.0 are not the ones with the fanciest tools, but the ones who move forward, learn as they go, and adapt in real time.

> **Key Takeaway** *Don't wait for permission — start small, prove value, and build momentum towards proactive safety.*

Chapter 8

Closing Thoughts and Next Steps



"Taking action is how you stay ahead"

Safety 4.0 isn't a buzzword. It's a mindset shift, a set of skills, and a toolkit that empowers you to move from firefighting to foresight. It's about making safety more human, more proactive, and more connected.

This eBook is just the beginning. If you want to go deeper, build confidence with the tools, future-proof your career and become the Safety 4.0 leader, you won't regret joining the Safety 4.0 Academy. It's where we take the ideas in this book and turn them into practice, with lessons, case studies, and step-by-step guidance you can use immediately.

Stay bold, stay ahead.

— Lucas Domingues

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